

SchoolEco — School Management System

Complete System Documentation

Version: 1.0

Institution: St. Dominic College

Platform URL: <http://localhost:3000>

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1. System Overview

SchoolEco is a **SaaS School Management System** designed as a connected "train system" where each administrative module is a station in a student's academic lifecycle. The system is built on the principle that data flows in one direction — from Admissions, through Registrar, Treasury, Academic Admin, Dean, Teacher, and finally to the Student — with each station dependent on the previous.

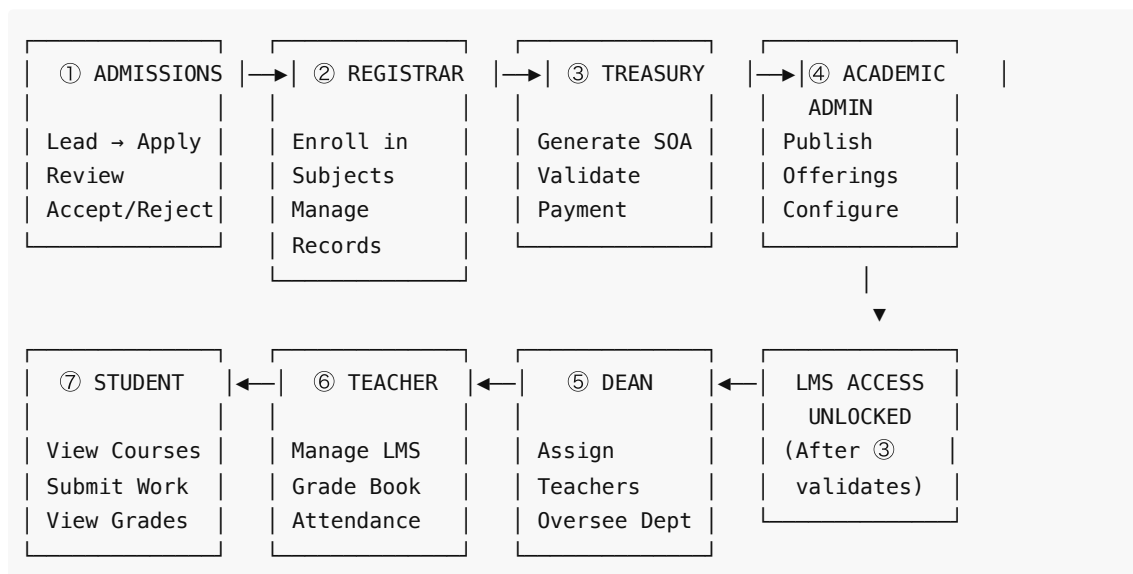
Core Philosophy

- **Single source of truth** — Each module owns its data. No duplication.
- **Sequential workflow** — A student cannot access the LMS until Treasury validates payment.
- **Role-based access** — Every user sees only what their role permits.
- **Non-breaking extensions** — New features are additive and never modify existing data flows.

What SchoolEco Manages

Domain	Scope
Admissions	Lead tracking, applicant intake, acceptance/rejection
Student Records	Personal data, academic history, enrollment status
Finance	Statement of Account, payments, cashier validation
Budget	Departmental budget allocation and tracking
Curriculum	Subjects, programs, course offerings, rooms
Faculty	Teacher accounts, subject assignments, schedules
LMS	Modules, assignments, quizzes, attendance, discussions
Grades	Grade entry, finalization pipeline, official records
Documents	Automated TOR, certificates, diploma generation

2. The Train System Pipeline



The Full Sequential Flow

1. **Admissions Officer** reviews an applicant and marks them **Accepted**
2. **Registrar** creates the official student record and **enrolls** the student in subjects
3. **Treasury** generates a Statement of Account and **validates payment**
4. **Academic Admin** publishes subject offerings and configures rooms
5. **Dean** reviews department and **assigns teachers** to published offerings
6. **Teacher** sets up their schedule, creates LMS content (modules, assignments, quizzes)
7. **Student** gains LMS access to enrolled subjects only after payment validation
8. **Teacher** grades submissions → **Grade Finalization Room** → **Registrar approves** → Official records updated

3. User Roles & Access

Role Hierarchy

SUPER_ADMIN
└─ ADMISSION_OFFICER
└─ REGISTRAR
└─ TREASURER
└─ ACCOUNTING
└─ ACADEMIC_ADMIN
└─ DEAN (4 departments)
└─ TEACHER
└─ STUDENT

Role Permissions Matrix

Feature	Super Admin	Admission	Registrar	Treasurer	Accounting	Academic
Accept/Reject Applicants	✓	✓	—	—	—	—
CRM Lead Management	✓	✓	—	—	—	—
Edit Student Records	✓	—	✓	—	—	—
Enroll Students	✓	—	✓	—	—	—
Generate/Validate SOA	✓	—	—	✓	—	—
Create Budget	✓	—	—	—	✓	—
View Dept Budget	✓	—	—	—	✓	—
Create Subjects/Offerings	✓	—	—	—	—	✓
Assign Teachers	✓	—	—	—	—	—
Manage LMS Content	✓	—	—	—	—	—
Submit Grades	✓	—	—	—	—	—
Approve Grades	✓	—	✓	—	—	—
View Own Grades	—	—	—	—	—	—
Generate Documents	✓	—	✓	—	—	—

Manage Custom Roles	✓	—	—	—	—	—
View Audit Logs	✓	—	—	—	—	—
Team Hub	✓	✓	✓	✓	✓	✓
Calendar	✓	✓	✓	✓	✓	✓

4. Demo Accounts

All demo passwords: **password**

Role	Email	Portal
Super Admin	admin@school.edu	/staff/dashboard
Admission Officer	admissions@school.edu	/staff/admissions
Registrar	registrar@school.edu	/staff/registrar
Treasurer	treasury@school.edu	/staff/treasury
Accounting	accounting@school.edu	/staff/treasury/budget
Academic Admin	academic@school.edu	/staff/academic
Dean — Computing	dean.computing@school.edu	/staff/dean
Dean — Business	dean.business@school.edu	/staff/dean
Dean — Nursing	dean.nursing@school.edu	/staff/dean
Dean — Arts	dean.arts@school.edu	/staff/dean
Teacher (Prof. Santos)	prof.santos@school.edu	/teacher/dashboard
Student (Ethan Dela Cruz)	student@school.edu	/student/dashboard

Public access: The applicant form at `/apply` requires no login.

5. Public Modules

5.1 Landing Page (/)

The root URL shows the public landing page for unauthenticated visitors. It includes:

- Hero section with system overview
- Pipeline cards showing the school management workflow
- Features grid
- Call-to-action to apply or contact the school
- Navigation links to Login and Apply

Authenticated users are automatically redirected to their portal.

5.2 Applicant Form (/apply)

Any prospective student can submit an application without logging in. Fields:

Personal Information

- First Name, Last Name, Middle Name
- Date of Birth, Gender
- Nationality, Religion
- Email Address, Phone Number
- Home Address

Academic Information

- Application Type: Freshman / Transferee / Returnee
- Previous School
- Previous Program (for transferees)
- Academic Year applying for

Supporting Documents

- Upload: Form 138 / Transcript, Birth Certificate, ID Photo
- File types: PDF, JPG, PNG, DOC, DOCX
- Simulated upload with 600ms progress indicator

Submission Flow:

1. Student fills out and submits form
 2. Application is saved to `sessionStorage` (survives page reload during login)
 3. Admissions officer finds the application in their queue with status **PENDING**
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6. Admissions Module

Access: ADMISSION_OFFICER, SUPER_ADMIN

URL: /staff/admissions

6.1 Applicant List

Displays all applicants with real-time status filtering:

- **All** — full list
- **Pending** — awaiting review
- **Under Review** — actively being assessed
- **Accepted** — approved for enrollment
- **Rejected** — not admitted

Features:

- Search is not exposed on the list; filtering is by status tab
- Clicking a row opens the full applicant detail page
- New applicants submitted via `/apply` appear automatically (rehydrated from `sessionStorage`)

6.2 Applicant Detail Page

Shows full applicant record with tabs:

Profile Tab

- All personal information from the application form
- Status badge with history
- Decision buttons: Accept / Reject (only ADMISSION_OFFICER and SUPER_ADMIN can decide)

Documents Tab

- Each required document shown as a row
- Upload functionality with status indicators (idle → uploading → success/error)
- Accepted file types: PDF, JPG, PNG, DOC, DOCX

Workflow

1. Admission Officer reviews the application
2. Clicks **Accept** → applicant status becomes ACCEPTED; a student record is created in the Registrar module
3. Clicks **Reject** → applicant status becomes REJECTED; a rejection reason is recorded

6.3 Admissions CRM (/staff/admissions/crm)

A full Kanban-style lead pipeline for prospective students who have not yet applied.

Pipeline Stages (left to right):

1. **New Lead** — Initial contact
2. **Contacted** — First outreach made
3. **Interested** — Expressed intent to apply
4. **Applicant** — Has submitted a formal application
5. **For Interview** — Invited for entrance interview
6. **Accepted** — Admitted to the school
7. **Enrolled** — Now an official student
8. **Lost** — No longer pursuing enrollment

CRM Features:

- Drag-and-drop cards between stages
- Slide-over detail panel with 3 tabs:
 - **Profile** — lead information
 - **Timeline** — activity log (auto-logged on stage change)
 - **Follow-ups** — scheduled calls, emails, interviews
- Log Activity button creates timeline entries
- Follow-up scheduler with due dates
- Auto-import of accepted applicants from the main admissions system

7. Registrar Module

Access: REGISTRAR, SUPER_ADMIN

URL: /staff/registrar

7.1 Student Records List

Displays all students in a table with:

- **Square avatar** showing student initials
- Student name and email
- Student ID (monospaced)

- Program code and name
- Year level
- Enrollment badge (ENROLLED, PRE_ENROLLED, etc.)
- Status badge (ACTIVE, INACTIVE, DROPPED, GRADUATED)
- "Manage →" link to the full record

Filters:

- Text search by name, ID, email, or program code
- Status filter pills: All / Active / Inactive / Graduated / Dropped
- Enrollment dropdown: All / Enrolled Only / Not Enrolled / Newly Added

Quick Drawer:

Clicking any student row opens a slide-over drawer showing:

- Large square photo area with initials (110×130px navy block)
- Name, email, student ID, status
- Personal Information table (name, DOB, gender, phone, address)
- Academic Information (program, year level, status)
- Current Enrollment (subjects enrolled this semester)
- **Account Balance** section:
 - If no SOA: greyed note
 - If fully paid: green confirmation with total
 - If balance pending: **red clickable card** showing balance amount and "Partial" badge
 - Clicking opens the **Balance Breakdown Modal**

Balance Breakdown Modal:

- Three-column summary: Total / Paid / Balance
- Itemized fee table (Tuition, Miscellaneous, Laboratory, etc.)
- Payment history with method, receipt number, and date
- Red outstanding balance callout

7.2 Student Detail Page (/staff/registrar/[studentId])

Four-tab record editor:

Tab 1: Personal Information

- All personal fields: Name, DOB, Place of Birth, Gender, Civil Status, Nationality, Religion, Blood Type
- Contact: Email (with verified toggle), Phone (with verified toggle)
- Address
- Student Status dropdown
- Edit mode toggle — non-editable fields are read-only for non-registrar roles

Tab 2: Family Background

- Addable family members: Name, Relation, Phone, Email, Occupation
- Add/remove rows only in edit mode

Tab 3: Academic Records

- Year Level dropdown (always visible, always editable by Registrar)
- Per-semester course table:
 - Subject code and name
 - Quiz Average, Assignment Average, Final Grade

- Grade Badge (PASSED, FAILED, INCOMPLETE, etc.)

Tab 4: Education History

- Previous schools with title, dates, and document upload
- Each entry supports a PDF/image upload
- Add/remove entries in edit mode

7.3 Document Automation System (/staff/registrar/documents)

See [Section 16](#) for full documentation.

7.4 Grade Finalization Room (/staff/grades)

See [Section 15](#) for full documentation.

8. Treasury Module

Access: TREASURER, SUPER_ADMIN

URL: /staff/treasury

8.1 Cashier (/staff/treasury)

The primary cashier interface for processing student payments.

Features:

- Search student by name or ID
- View current Statement of Account (SOA)
- Record payment: amount, method (Cash / GCash / Online / Bank), reference number
- Print/download receipt
- SOA status updates automatically: UNPAID → PARTIAL → PAID

8.2 Student Accounts (/staff/treasury/accounts)

Overview of all student financial accounts:

- SOA status per student
- Total balance outstanding
- Payment history

8.3 Transaction Logs (/staff/treasury/logs)

Append-only log of all financial transactions:

- Payment date, amount, method
- Receipt/reference numbers
- Validated by (staff name)
- SOA linked

8.4 Statement of Account (SOA) Generation

When a student is enrolled, the Registrar triggers SOA generation:

1. Treasury creates SOA items based on the program's fee schedule
2. Items include: Tuition Fee, Miscellaneous Fee, Laboratory Fee, Registration Fee, ID Fee, NSTP Fee
3. SOA status begins as **UNPAID**

4. As payments are recorded and validated, status moves to **PARTIAL** then **PAID**
5. **PAID** status triggers automatic LMS access for the student

SOA Statuses:

Status	Meaning
UNPAID	No payment recorded
PARTIAL	Some payment made, balance remains
PAID	Fully settled
OVERPAID	Payment exceeds total (refund may be needed)

9. Accounting / Budget Module

Access: ACCOUNTING, SUPER_ADMIN

URL: /staff/treasury/budget

See [Section 18](#) for full documentation.

10. Academic Admin Module

Access: ACADEMIC_ADMIN, SUPER_ADMIN

URL: /staff/academic

10.1 Subject Master List (/staff/academic)

Manages the school's subject catalog.

Fields per subject:

- Subject Code (e.g., CS101)
- Subject Name
- Lecture Units
- Lab Units
- Total Hours (auto-calculated: (Lecture + Lab) × 18)
- Type (Core, Elective, GE, etc.)
- Description
- Eligibility Conditions (optional, collapsible):
 - Prerequisite subjects
 - Minimum units completed
 - Year level requirement
 - Minimum grade from previous subject
 - Program restriction

10.2 Subject Offerings (/staff/academic/offerings)

Manages which subjects are offered each semester.

Fields per offering:

- Subject (from master list)

- Section code (e.g., BSCS-1A)
- Semester
- Capacity
- Status: DRAFT → PUBLISHED → ARCHIVED

Workflow:

1. Academic Admin creates an offering for the active semester
2. Offering starts as DRAFT — no teacher assigned, no schedule
3. Dean assigns a teacher → offering moves toward PUBLISHED
4. Teacher sets schedule → offering is fully PUBLISHED
5. Students can be enrolled in PUBLISHED offerings only

Note: Start Time / End Time are NOT set in the offering. Schedule is set by the teacher after the Dean assigns them.

10.3 Departments (/staff/academic/departments)

Manages academic departments and their programs:

- Department name and code
- College affiliation
- Programs list (e.g., BSCS, BSIT under Computing)

10.4 Rooms (/staff/academic/rooms)

Manages physical spaces:

- Room name and capacity
- Room type (Lecture, Laboratory, etc.)
- Availability windows (set by Academic Admin; teachers schedule within these windows)

10.5 School Year Calendar (/staff/calendar)

See [Section 20](#) for full documentation.

11. Dean Module

Access: DEAN (scoped to their department), SUPER_ADMIN

URL: /staff/dean

Each Dean account has a `deanDepartment` field (e.g., "College of Computing") that scopes all data they can see.

11.1 Dean Dashboard

Animated statistics dashboard showing:

- Enrollment count for the department
- Active students by year level
- Recharts bar chart of enrollment trends with school year filter
- Year-level accordion breakdown

11.2 Programs (/staff/dean/programs)

Read-only view of the department's programs:

- Program code and name
- Number of enrolled students per program

11.3 Student List (/staff/dean/students)

Read-only list of all students in the department:

- Filter by year level and enrollment status
- ?year=N URL parameter pre-filters to a specific year
- Cannot edit student records (Registrar only)

11.4 Teacher Assignment (/staff/dean/assignments)

This is the most critical Dean function.

Two tabs:

- **Needs Teacher** — Published offerings with no teacher assigned
- **Assigned** — Offerings with confirmed teacher assignments

Assignment workflow:

1. Dean sees a published offering in "Needs Teacher"
2. Selects a faculty member from the department
3. Clicks Assign → TeacherAssignment record created
4. Offering now appears in teacher's "My Courses" page
5. Teacher can now set their schedule for this offering

11.5 Department Budget (/staff/dean/budget)

Read-only. Shows:

- Summary cards: Total Allocated / Total Used / Remaining
 - Per-budget health card with progress bar
 - Itemized expense list per budget
 - Deans cannot create or edit budgets
-

12. Teacher / Faculty Module

Access: TEACHER

URL: /teacher/dashboard

12.1 Teacher Dashboard

Overview showing:

- Assigned courses count
- Pending submissions count
- Pending grading count
- Quick links to active courses

12.2 My Courses (/teacher/subjects)

Card grid of all assigned and published offerings:

- Subject name and code
- Section and semester

- Enrollment count
- Schedule days/times
- Quick links to sub-pages

12.3 Course Detail Page (/teacher/subjects/[offeringId])

Nine navigation cards leading to all LMS management areas:

Card	Purpose
Learning Materials	Upload modules, PDFs, videos, links
Assignments	Create, publish, grade assignments
Quizzes & Exams	Create, schedule, auto-grade quizzes
Grade Book	Compute and submit final grades
Attendance	Record and track class attendance
Announcements	Post course-wide announcements
Discussions	Monitor and respond to discussions
Grading Criteria	Set grade weights and passing threshold
Student List	View enrolled students

12.4 Schedule Management (/teacher/schedule)

Access gate: Only shows offerings that are PUBLISHED and assigned to this teacher.

- View available time slots
- Add schedule entries for assigned offerings
- Room selection (validated against `MOCK_ROOM_AVAILABILITY`)
- Conflict detection: same room, overlapping time → error

12.5 Grade Finalization Flow

1. Teacher records grades in the Grade Book
2. Clicks "Finalize Grades" → creates `GradeSubmission` → offering is locked
3. Registrar reviews in the Grade Finalization Room
4. On approval: grades become official records, students can see them
5. On rejection: lock removed, teacher corrects and resubmits

13. LMS — Learning Management System

The LMS is not a separate application. It is an **academic extension** of SchoolEco, consuming and synchronizing data from existing modules. It does not create its own student, faculty, or subject records.

13.1 LMS Access Control

A student gains LMS access to a subject only if **all three conditions are true**:

1. The student is **officially enrolled** in the offering
2. The enrollment status is **ENROLLED** (Treasury has validated payment)

3. The subject offering is **PUBLISHED**

If payment is pending (`PRE_ENROLLED` status), the student sees a **locked screen** explaining that access is pending payment validation.

13.2 Auto-Synchronization

Source Module	Data Synchronized to LMS
Registrar + Enrollment	Student name, ID, enrolled subjects
Academic Admin	Subject code, name, units, section
Dean Module	Faculty assignment to each offering
Treasury	Payment status → LMS access unlock

No manual account creation. LMS users are the same users from the existing authentication system.

13.3 Student LMS Experience

Subject List (`/student/subjects`)

- Shows all enrolled subjects as cards
- Each card shows subject name, code, schedule, and enrollment status badge
- Links to individual course pages

Course Page (`/student/subjects/[offeringId]`)

- **Access Gate:** If not ENROLLED → full-screen lock explaining payment pending
- **Pinned Announcement Banner** — shows the instructor's most recent pinned announcement
- **Upcoming Deadlines Row** — assignments and quizzes due within the semester
- **7-Card Navigation Grid:**
 - Announcements (with count)
 - Learning Materials (with file count)
 - Assignments (with submission status)
 - Quizzes & Exams (with attempt count)
 - Discussions (with reply count)
 - My Attendance (shows attendance percentage, color-coded)
 - Grade Summary (shows current final grade)
- **Grade Summary Card** — quiz avg, assignment avg, midterm, final grade
- **Attendance Summary** — Present / Late / Absent / Excused counts with progress bar and warning if below 75%

Announcements (`/student/subjects/[offeringId]/announcements`)

- Read-only list sorted pinned-first, then by date
- Pinned announcements show a pin icon
- Teacher name and date shown per announcement

Discussions (`/student/subjects/[offeringId]/discussions`)

- Accordion-style thread list
- Click to expand and see all replies
- Students can post replies (inline input, Enter to submit)
- Teacher replies are marked with an "Instructor" badge

- Pinned posts appear first

Learning Materials (/student/subjects/[offeringId]/materials)

- Organized by modules
- Supports: PDF, Video, Link, File
- Module title and description shown
- Published modules only (draft modules hidden)

Assignments (/student/subjects/[offeringId]/assignments)

- List of published assignments with due dates
- Submit via text input or file upload
- Submission status: Not Submitted / Submitted / Graded
- Grade and feedback shown after grading

Quizzes (/student/subjects/[offeringId]/quizzes)

- Shows available quizzes with time window
- Attempt history and score
- Supports MCQ, True/False, Essay questions

13.4 Teacher LMS Management

Learning Materials (/teacher/subjects/[offeringId]/materials)

- Create and publish modules
- Add materials to each module (PDF, Video, Link, File)
- Toggle module publish/draft status
- Materials only visible to students when module is published

Assignments

- Create assignments with title, description, due date, total points
- Publish/unpublish control
- View all student submissions
- Grade with numeric score and written feedback

Quizzes & Exams

- Create quizzes with time limit and date window
- Add questions: MCQ (up to 4 options), True/False, Essay
- Set points per question
- View attempt results per student

Grade Book (/teacher/subjects/[offeringId]/grades)

- Table of all enrolled students
- Enter: Quiz Average, Assignment Average, Midterm Grade, Final Grade
- Grade formula displayed (based on Grading Criteria)
- "Finalize Grades" button → submits to Grade Finalization Room
- After finalization: offering is LOCKED (editing disabled unless rejected)

Grading Criteria (/teacher/subjects/[offeringId]/criteria)

- Range sliders + number inputs for grade weights
- Components: Quiz, Assignment, Midterm Exam, Final Exam
- Weights must sum to 100%

- Passing grade threshold setting (default: 75)
- Live formula preview
- Saved to `MOCK_GRADE_CRITERIA`

Attendance (`/teacher/subjects/[offeringId]/attendance`)

Record Tab:

- Date selector (defaults to today, max = today)
- Student list with per-student status buttons: Present / Late / Absent / Excused
- Running totals shown (P: X, L: X, A: X)
- "Save Attendance" button confirms to database

History Tab:

- Dot-matrix table: each column = a date, each row = a student
- Color dots: Green = Present, Amber = Late, Red = Absent, Blue = Excused, Grey = No record
- Attendance rate % per student (color-coded)

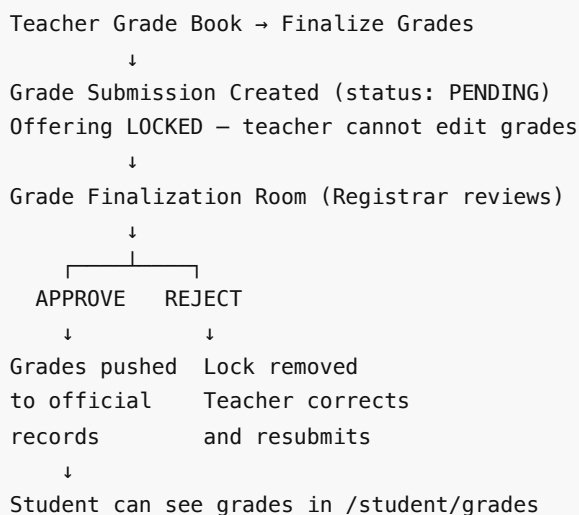
Announcements (`/teacher/subjects/[offeringId]/announcements`)

- Create announcement with title, content, optional pin
- Pinned announcements appear first and are highlighted in red
- Delete any announcement
- Students see all announcements in real-time

Discussions (`/teacher/subjects/[offeringId]/discussions`)

- View all student-created and teacher-created posts
- Pin/unpin posts for prominence
- Reply to any post with "Instructor" badge
- Monitor and moderate all discussion activity

13.5 LMS Grade Flow (Critical Path)



Grades inside the LMS are NOT automatically official. They only become part of the student's permanent record after the Registrar approves them in the Grade Finalization Room.

14. Student Portal

Access: STUDENT

URL: /student/dashboard

14.1 Dashboard

Overview of the student's current semester:

- Enrollment status
- Subjects enrolled this semester
- SOA balance and payment status
- Recent grades
- Notifications

14.2 Enrollment (/student/enrollment)

Read-only view of enrolled subjects:

- Subject name, code, units
- Section and schedule
- Faculty name
- Enrollment status badge per subject

14.3 My Subjects (/student/subjects)

Card-style list of all enrolled subjects linking to individual LMS course pages. Each card shows:

- Subject name and code
- Schedule (day/time)
- Faculty name
- Enrollment status badge

14.4 My Grades (/student/grades)

Only shows grades from APPROVED submissions — pending or rejected submissions are never displayed.

Table columns:

- Subject name and code
- Units
- Final Grade (numeric)
- Rating (letter grade)
- Status badge (PASSED / FAILED)
- Approval date

Empty state shown if no grades have been officially approved yet.

14.5 Statement of Account (/student/soa)

Detailed financial view:

- SOA status (UNPAID / PARTIAL / PAID / OVERPAID)
- Itemized fee breakdown
- Payment history with receipt numbers
- Outstanding balance

14.6 Notifications (/student/notifications)

System-generated notifications for:

- New assignments posted
- Grades released
- Upcoming deadlines
- LMS announcements
- Payment confirmations

14.7 My Profile (/student/profile)

Student can view (but not edit) their profile information. Photo can be changed using the crop editor.

15. Grade Finalization Room

Access: REGISTRAR, ACADEMIC_ADMIN, SUPER_ADMIN

URL: /staff/grades

This is the official grade approval workflow.

15.1 Three Tabs

Pending Tab

- Lists all grade submissions awaiting review
- Shows: offering name, section, faculty, submission date, number of students
- Click "Review" to open the detailed grade table

Approved Tab

- Historical log of approved grade batches
- Shows approval date and approving staff

Rejected Tab

- Grade batches returned to the teacher
- Shows rejection reason and return date

15.2 Review Modal

When reviewing a grade submission:

- Full table of all students with their numeric grades
- Grade status per student (PASSED / FAILED)
- Two actions:
 - **Approve** → grades are pushed to `MOCK_GRADES` (official records), audit log created, student notification sent
 - **Reject** → reason is required, teacher receives notification, offering is unlocked for correction

15.3 Audit Trail

Every grade finalization action (approve or reject) creates an entry in the audit log with:

- Action type

- Actor (who approved/rejected)
- Timestamp
- Subject offering and semester

16. Document Automation System

Access: REGISTRAR, SUPER_ADMIN

URL: /staff/registrar/documents

A full document generation engine supporting three template modes.

16.1 Three Template Types

1. SmartDocs Builder (DOCX from scratch)

- Rich text editor canvas
- Right-side Fields Panel with:
 - Personal fields (Full Name, Student ID, DOB, etc.)
 - Academic fields (Program, Year Level, GWA, Units)
 - Auto-fill Tables (subjects loop)
 - Show/Hide Conditions (based on student data)
- Drag-and-drop field insertion
- Smart Table Builder: choose loop type (all subjects / current / completed) and columns
- Conditional blocks: `{{#if is_graduated}}...{{/if}}`

2. DOCX Field Mapping (Upload existing DOCX)

- Upload any Word document
- Click text areas to map them to student fields
- Click table rows to configure as data loops
- Preview with actual student data

3. PDF Overlay Editor (Upload PDF form)

- Upload a PDF template (e.g., official form, certificate)
- Visual canvas with the PDF rendered as a background image (via pdf-lib/pdfjs)
- Tool modes: Select, Text, Number, Date, Image, Table
- Click to place field overlays anywhere on the PDF
- Drag to move, corner-handle to resize
- Right panel: map overlay to student field, set font size, bold, color, alignment
- Preview generated PDF with real student data

16.2 Template Engine Pipeline

When generating a document, the system processes templates in this order:

1. **Loop expansion** — `<tr data-loop-type="subjects">` rows cloned per subject
2. **Special tokens** — condition and placeholder spans converted to raw `{{tokens}}`
3. **Conditional blocks** — `{{#if cond}}...{{else}}...{{/if}}` evaluated per student
4. **Filtered tokens** — `{{key | filter}}` processed (uppercase, lowercase, date format, etc.)
5. **Simple substitution** — remaining `{{key}}` replaced with student data

Supported Conditions: `is_graduated` , `is_active` , `is_dropped` , `is_honor_student` , `is_male` , `is_female` , `has_subjects` , `has_gwa`

Supported Filters: uppercase , lowercase , title , or (fallback), number (decimals), format (date)

16.3 Generate Tab (3-Step Wizard)

Step 1: Select a template

Step 2: Select student(s) — searchable table, selected student panel shown above

Step 3: Preview the generated document

Step 4: Download or batch generate

Step 5: Document is logged in History tab

16.4 Document History

All generated documents are logged with:

- Template name
 - Student name
 - Generated by (staff)
 - Date and time
 - Download link
-

17. User Management

Access: SUPER_ADMIN only

URL: /staff/users

17.1 Users Tab

Read-only table of all system accounts:

- Name with square avatar
- Email address
- Role badge (color-coded per role)
- Status (Active / Inactive)
- Account type: "System / Locked" for built-in accounts

17.2 Role Management Tab

System Roles (Read-Only)

Eight locked roles displayed as cards with a padlock icon:

- Super Admin, Admission Officer, Registrar, Treasurer, Accounting, Academic Admin, Dean, Teacher, Student
- Cannot be edited, deleted, or modified

Custom Roles (Editable)

Super Admin can create unlimited custom roles:

Create Role Modal:

- Role Name (required)
- Description (optional)
- Permission Matrix: 7 modules × 4 permission levels

Module	View	Create	Edit	Delete
Admissions	☐	☐	☐	☐
Academic Admin	☐	☐	☐	☐
Registrar	☐	☐	☐	☐
Treasury	☐	☐	☐	☐
LMS	☐	☐	☐	☐
Reports	☐	☐	☐	☐
User Management	☐	☐	☐	☐

- Enabling Create / Edit / Delete automatically enables View
- Clicking the module name toggles all four permissions at once
- Custom roles can be edited or deleted anytime
- Deleted roles do not affect existing users (display fallback)

18. Budget Management

Access: ACCOUNTING (create/edit), SUPER_ADMIN (full), DEAN (view-own, read-only)

Accounting URL: /staff/treasury/budget

Dean URL: /staff/dean/budget

18.1 Key Principle

The Budget Management module is **completely isolated** from Treasury. It does not touch SOA, payments, or student billing. It tracks departmental operational budgets only.

18.2 Departments

Budgets are allocated to the four academic colleges:

- College of Computing
- College of Business
- College of Nursing
- Arts & Sciences

18.3 Overview Tab

- **4 Summary Cards:** Total Allocated / Total Used / Total Remaining / Budget Alerts
- **Budget Health Cards** (one per filtered department):
 - Budget name and department
 - Health indicator icon (✓ On Track / ⚠ Near Limit / ✗ Depleted)
 - Three amount boxes: Total / Used / Remaining
 - Progress bar (green < 80%, amber 80–99%, red ≥ 100%)
 - Period type and date range
 - Edit and delete buttons

18.4 Budgets Tab

Table view of all budgets with:

- Budget name
- Department
- Total, Used, Remaining amounts
- Period type badge (Monthly / Quarterly / Yearly)
- Inline progress bar with percentage
- Edit / Delete actions

18.5 Create Budget Form

Field	Required	Notes
Budget Name	Yes	Free text
Department	Yes	Dropdown: 4 colleges
Budget Amount (₱)	Yes	Must be > 0
Period Type	Yes	Monthly / Quarterly / Yearly
Start Date	Yes	
End Date	Yes	Must be after Start Date

18.6 Expenses Tab

Chronological log of all recorded expenses:

- Description
- Department
- Linked budget name
- Amount (shown as negative / deducted)
- Date
- Recorded by (staff name)

18.7 Record Expense Form

Field	Required	Notes
Budget	Yes	Dropdown showing remaining balance live
Description	Yes	Free text
Amount (₱)	Yes	Cannot exceed remaining balance
Date	Yes	

18.8 Department Budget View (Dean)

Deans see a read-only version for their own college:

- Summary cards
- Per-budget card with itemized expenses
- Progress bars with health indicator
- Clear warning messages when near limit or depleted
- No create, edit, or delete buttons

18.9 Budget Health Rules

Usage	Status	Color
0–79%	On Track	Green
80–99%	Near Limit	Amber + warning message
≥ 100%	Depleted	Red + critical message

19. Team Hub

Access: All staff and teacher roles

URL: /staff/team , /teacher/team

19.1 Card Grid View

All school staff displayed as square cards in a responsive grid (2 → 5 columns):

- **Square photo** (80×80px, rounded-xl): Shows uploaded photo or initials
- Role badge (color-coded)
- Department, email, phone

19.2 Filters

- **Role Tabs:** All / Admin / Deans / Faculty (pill buttons, no stat counts)
- **Department Dropdown:** Filter by specific department

19.3 Profile Card Modal

Clicking any staff card opens a modal with:

- Profile photo (clickable to trigger photo editor)
- Full Name, Role badge
- "Add Photo" / "Edit Photo" button
- Details table: Full Name, Job Title, Department, Email, Phone, Birthday

19.4 Photo Upload & Crop Editor

A canvas-based photo editor appears when editing a profile photo:

Features:

- **280×280 circular crop canvas** — dark overlay shows exactly what gets saved
 - **Drag to pan** — mouse or touch to reposition the image within the crop circle
 - **Zoom controls:**
 - Scroll wheel (on desktop)
 - Pinch gesture (on mobile)
 - Range slider (explicit control)
 - +/- buttons
 - **Minimum zoom** — image always covers the full crop circle (no gaps)
 - **Upload Photo** button — opens file picker (accepts all image formats)
 - **Save Photo** — exports a 200×200 JPEG, saves to member record
 - Photo appears immediately on the card grid after saving
-

20. School Year Calendar

Access: All staff and teacher roles (view); ACADEMIC_ADMIN + SUPER_ADMIN (create/edit/delete)

URL: /staff/calendar , /teacher/calendar

20.1 Calendar Views

- **Monthly Grid** — standard calendar grid showing all events
- **Upcoming Events Sidebar** — chronological list of next 5 events
- **Full Event Table** — all events in a sortable table

20.2 Event Types and Colors

Type	Description	Color
HOLIDAY	School-wide non-working days	Red
ACADEMIC	Semester dates, academic milestones	Blue
ENROLLMENT	Enrollment periods	Green
EXAM	Midterm and final exam periods	Amber
ACTIVITY	School events, activities, sports fests	Purple
ADMIN	Administrative deadlines	Slate

20.3 Legend / Filter Toggles

Clicking a legend badge toggles that event type on/off in both the grid and upcoming sidebar.

20.4 Event Management (Academic Admin / Super Admin only)

- **Add Event** button → modal with: title, date range, type, description
- **Edit** (pencil icon on event) → same modal pre-filled
- **Delete** (trash icon) → confirmation dialog

21. Audit Logs

Access: SUPER_ADMIN only

URL: /staff/audit

Append-only log of all significant system actions:

- Grade submissions, approvals, rejections
- Student status changes
- Document generation
- Payment validations

Each entry shows:

- Timestamp
- Actor (who performed the action)
- Action type
- Affected entity (student name, subject, etc.)

- Details

Logs are read-only and cannot be deleted.

22. Notifications

22.1 System-Generated Notifications

Trigger	Recipients
Grade submission	Registrar
Grade approved	Teacher + Students
Grade rejected	Teacher
New assignment posted	Students (enrolled in that offering)
Payment validated	Student
New announcement	Students (enrolled in that offering)

22.2 Notification Bell (Header)

- Bell icon in the top header shows unread count badge
 - Clicking opens a notification dropdown
 - Notifications are marked read on view
-

23. Profile & Personalization

23.1 My Profile (`/staff/profile` , `/teacher/subjects`)

Editable staff profile:

- Profile photo (upload from device)
- Nickname / Display Name
- Phone number
- Job Title
- Birthday
- All saved to `localStorage` (persists across sessions)

23.2 Personalization (`/staff/personalization`)

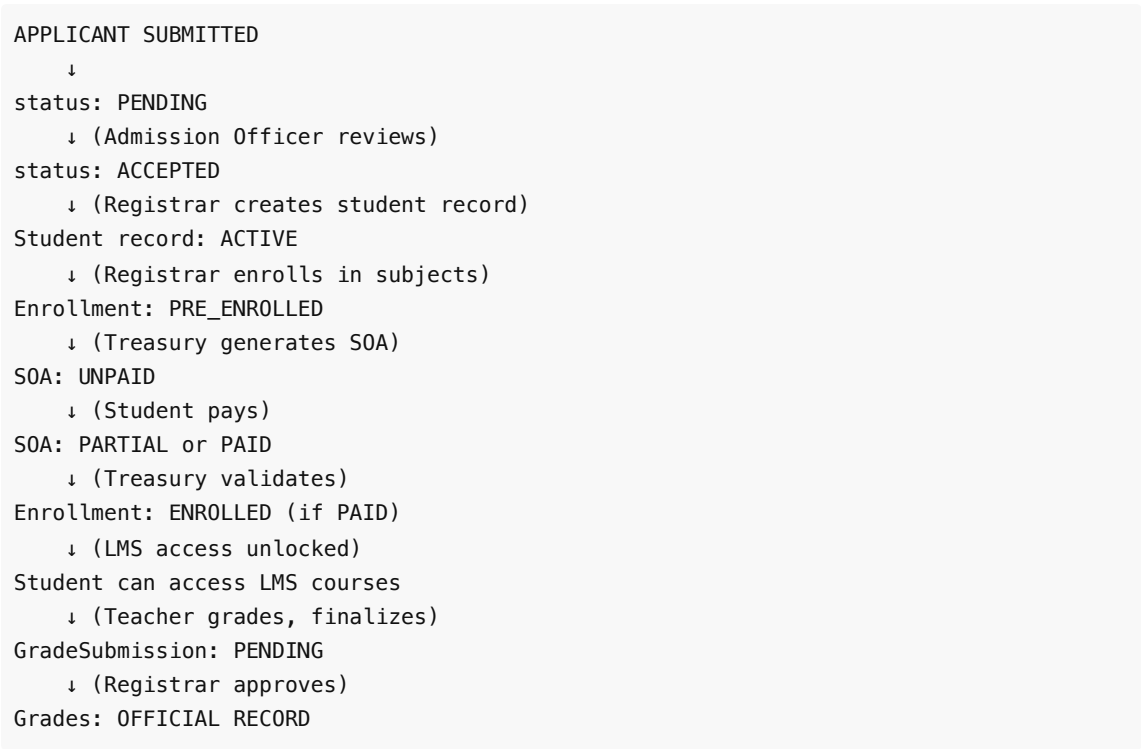
- **Theme toggle:** Light / Dark mode
- **Accent color:** Choose from preset brand colors
- Saved to `localStorage`

23.3 Settings (`/staff/settings`)

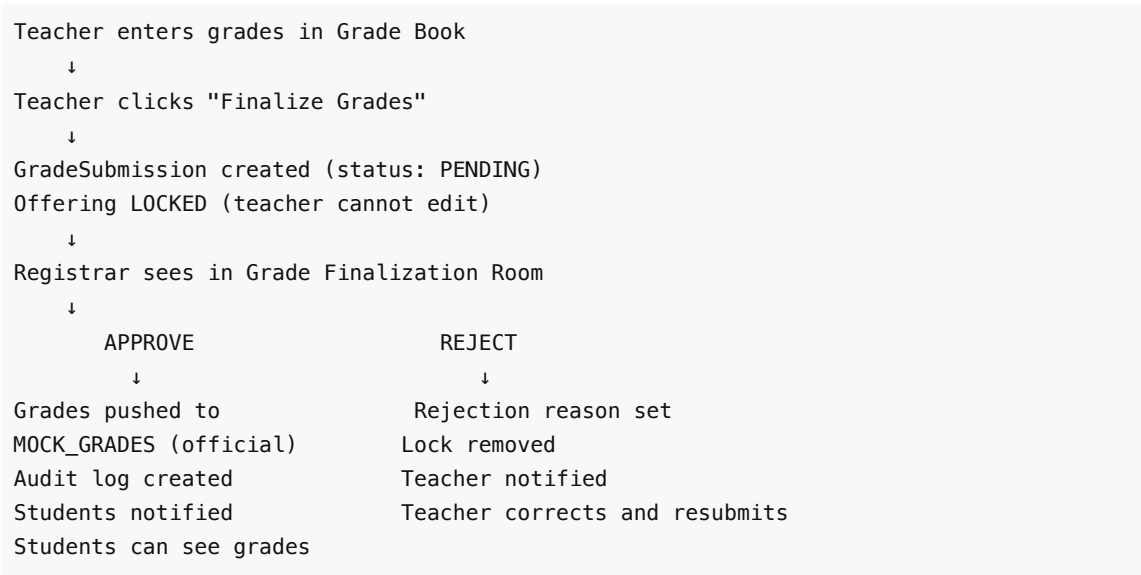
- Password change form
 - Notification preferences (toggle per notification type)
-

24. Complete Data Flow Reference

Student Lifecycle

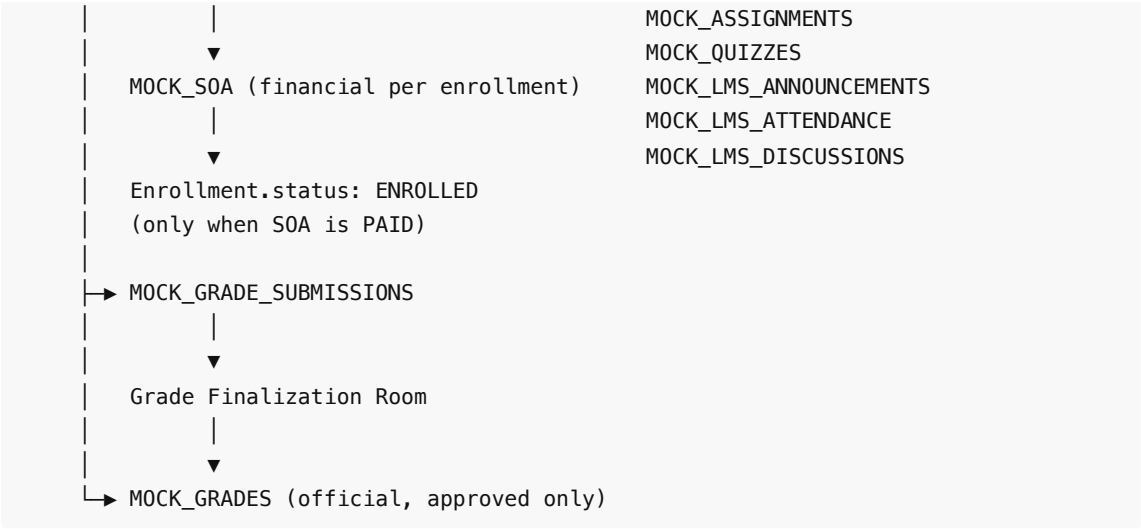


Grade Flow



Module Dependency Map





25. Technical Architecture

Technology Stack

Layer	Technology
Framework	Next.js 14 (App Router)
Language	TypeScript
Styling	Tailwind CSS
Database ORM	Prisma (SQLite in dev)
Authentication	NextAuth.js (JWT strategy)
Charts	Recharts
PDF Generation	pdf-lib
PDF Rendering	pdfjs-dist
Form Handling	react-hook-form + Zod
Icons	Lucide React

Project Structure

```
Application/
├── src/
│   ├── app/
│   │   ├── (auth)/login/           # Login page
│   │   ├── (public)/apply/         # Applicant form
│   │   ├── staff/                  # All staff portals
│   │   │   ├── admissions/         # + /crm
│   │   │   ├── registrar/         # + /[studentId] + /documents + /grades
│   │   │   └── treasury/          # + /accounts + /logs + /budget
```

```

| | | | | └─ academic/          # + /departments + /rooms + /offerings
| | | | | └─ dean/            # + /programs + /students + /assignments + /budget
| | | | | └─ grades/          # Grade Finalization Room
| | | | | └─ users/           # User Management
| | | | | └─ team/            # Team Hub
| | | | | └─ calendar/        # School Year Calendar
| | | | | └─ audit/           # Audit Logs
| | | | | └─ profile/         # Staff Profile
| | | | | └─ personalization/ # Theme/Accent
| | | | | └─ settings/        # Password/Notifications
| | | | | └─ help/           # Help Module
| | | └─ teacher/
| | | | └─ dashboard/
| | | | └─ subjects/[offeringId]/
| | | | | └─ materials/
| | | | | └─ assignments/
| | | | | └─ quizzes/
| | | | | └─ grades/
| | | | | └─ attendance/
| | | | | └─ announcements/
| | | | | └─ discussions/
| | | | | └─ criteria/
| | | | | └─ students/
| | | | └─ schedule/
| | | | └─ team/
| | | | └─ calendar/
| | └─ student/
| | | └─ dashboard/
| | | └─ subjects/[offeringId]/
| | | | └─ materials/
| | | | └─ assignments/
| | | | └─ quizzes/
| | | | └─ announcements/
| | | | └─ discussions/
| | | └─ grades/
| | | └─ soa/
| | | └─ enrollment/
| | | └─ notifications/
| | | └─ profile/
| └─ components/
| | └─ layout/          # Sidebars, Header
| | └─ ui/              # Badge, Button, Card, Table, Modal, Input, Avatar
| | └─ landing/         # Public landing page
| | └─ shared/          # ProcessFlow, etc.
| └─ lib/
| | └─ auth.ts          # NextAuth options + DEMO_USERS
| | └─ mock-data.ts     # All in-memory data + LMS seed
| | └─ utils.ts         # Helpers: cn(), fullName(), ROLE_LABELS,
ROLE_PORTALS
| | └─ db.ts            # Prisma client singleton
| | └─ help-registry.ts # Help module feature registry
| └─ types/index.ts     # All TypeScript interfaces

```

└─ prisma/	
└─┬─ schema.prisma	# Full relational schema (20+ models)
└─┬─ seed.ts	# Database seed script
└─ CLAUDE.md	# Developer guide

Authentication & Session

- **Auth provider:** NextAuth.js with Credentials provider
- **Session strategy:** JWT
- **Demo accounts:** Plain-text password comparison (not bcrypt) for fast load
- **JWT payload:** id , role , schoolId , schoolName , schoolColor , deanDepartment (Dean only), facultyId (Teacher only), studentId (Student only)
- **Session validation:** Each layout (staff/layout.tsx , teacher/layout.tsx , student/layout.tsx) calls getSession and redirects on mismatch

Design System

Primary Color: Navy Blue (#1a4a8a = brand-500) **Font:** Plus Jakarta Sans (primary), Playfair Display (formal/display), JetBrains Mono (code/IDs)

Color Tokens:

Token	Hex	Usage
brand-500	#1a4a8a	Primary buttons, active states
brand-600	#163d73	Hover
brand-900	#09182e	Super Admin badge, dark headers
gold-500	#c9a224	Premium accents
sidebar-bg	#0c1e3d	All sidebar backgrounds

Component Conventions:

- <SectionTitle> — page heading with 3px navy accent bar
- <Card padding="none"> — wrap tables with zero inner padding
- <Badge> — always use typed variants (RoleBadge, EnrollmentBadge, etc.)
- <Modal> — consistent header + footer structure
- Sign-out confirmation — all 3 sidebars show a confirmation dialog (never one-click)

Data Persistence Model

Scope	Mechanism	Resets On
Applicant form → Admissions	sessionStorage	Browser close
LMS seed data (offerings, modules, etc.)	Module-level IIFE in mock-data.ts	Hard reload
Budget data	Module-level arrays	Hard reload

Document templates and history	Module-level arrays in documents/page.tsx	Hard reload
Custom roles	MOCK_CUSTOM_ROLES array	Hard reload
Team Hub photos	member.avatar mutation	Hard reload
User Profile / Theme	localStorage	Never
All other mock data	Module-level arrays	Hard reload

z-Index Hierarchy

Element	z-index
Header (sticky)	35
CRM detail slide-over	36
Sidebars	40
Modal backdrop + dialog	50
Sign-out confirmation	200
Balance breakdown (stacked)	300

Appendix: Full Test Walkthrough

To test the complete end-to-end workflow from a fresh state:

Step 1: Submit Application (Anonymous)

1. Go to `/apply` (no login needed)
2. Fill out applicant form and submit
3. Application saved with status PENDING

Step 2: Admissions Review

1. Login as `admissions@school.edu`
2. Go to Applicants list — find the new application
3. Click to review full profile
4. Click **Accept** → student record created

Step 3: Registrar Enrollment

1. Login as `registrar@school.edu`
2. Find the new student in Student Records
3. Open their record → enroll in subjects (CS101, CS201)
4. Enrollment status: PRE_ENROLLED

Step 4: Treasury Payment

1. Login as `treasury@school.edu`
2. Find student in Cashier
3. Generate SOA → items appear (Tuition, Fees, etc.)

4. Record payment → validate → SOA status: PAID
5. Enrollment status automatically updates to ENROLLED

Step 5: Academic Admin Setup

1. Login as `academic@school.edu`
2. Verify offerings are PUBLISHED
3. Configure rooms if needed

Step 6: Dean Assigns Teacher

1. Login as `dean.computing@school.edu`
2. Go to Teacher Assignment
3. Assign Prof. Santos to the offerings

Step 7: Teacher Prepares LMS

1. Login as `prof.santos@school.edu`
2. Go to My Courses → click the offering
3. Add module with materials
4. Create assignment and quiz
5. Post announcement

Step 8: Student Accesses LMS

1. Login as `student@school.edu`
2. Go to My Subjects → click CS101
3. LMS is now active (payment validated)
4. View materials, submit assignment, take quiz

Step 9: Teacher Grades and Finalizes

1. Login as `prof.santos@school.edu`
2. Go to Grade Book → enter grades
3. Click "Finalize Grades" → offering is locked

Step 10: Registrar Approves Grades

1. Login as `registrar@school.edu`
2. Go to Grade Finalization Room → Pending tab
3. Review grade table → click Approve
4. Grades are now official

Step 11: Student Sees Official Grades

1. Login as `student@school.edu`
2. Go to My Grades
3. Official approved grades are now visible